| 1 | 2 | 3 | 4 | 5 | 6 | Optional | Sum |
|---|---|---|---|---|---|---|---|
| /20 | /20 | /20 | /20 | /20 | /20 | /20 | /100 |

Name ________________________ Dec. 24, 2003 Wednesday

Department __________________

SID _________________________

***University Physics***

Sample Problems for Final Examination

School of Intensive Instruction in Sciences and Arts, Nanjing University

Physical Constants

| | |
|---|---|
| Electron charge magnitude | $e = 1.60\times10^{-19}$ C |
| Dielectric constant of vacuum | $\varepsilon_0 = 8.85\times10^{-12}$ F/m |
| Permeability of vacuum | $\mu_0 = 4\pi\times10^{-7}$ N/A$^2$ |
| Vacuum speed of light | $c = 3.00\times10^{8}$ m/s |
| Electron rest mass | $m_e = 9.11\times10^{-31}$ kg $= 0.511$Mev/c$^2$ |
| Proton rest mass | $m_p = 1.672\times10^{-27}$ kg $= 938.3$MeV/c$^2$ |
| Neutron rest mass | $m_n = 1.675\times10^{-27}$ kg $= 939.6$ MeV/c$^2$ |
| Atomic mass unit | 1 amu $= 1.660\times10^{-27}$ kg $= 931.5$MeV/c$^2$ |
| Electron volt (eV) | 1eV=$1.60\times10^{-19}$ J |
| Planck constant | $h = 6.63\times10^{-31}$ Js $\hbar = 2.11\times10^{-31}$ Js $= 0.66\times10^{-15}$ eVs |
| Boltzmann constant | $k_B = 1.38\times10^{-23}$ J/K $= 8.62\times10^{-5}$ eV/K |
| Bohr radius | $a_0 = 0.259$ Angstroms |

Select five out of following six problems.

1. (20pts). (a) In Compton scattering, find the maximum increase of the wavelength of the photon. (b) If the wavelength of the scattered photon is the twice of the wavelength of the incident photon, what is the minimum energy of the incident X-ray?

2.(20 pts) (a) Write the time-dependent Schrödinger equation and the stationary (time independent) Schrödinger equation for a particle of mass $m$ moving in the one-dimensional harmonic potential

$$U(x) = \frac{1}{2}kx^2$$

(b) The solution for the stationary Schrödinger equation in the one-dimensional harmonic potential is

$$E_n = (n + \tfrac{1}{2})\hbar\omega, \quad \psi_n(x) = N_n e^{-\frac{\alpha^2 x^2}{2}} H_n(\alpha x), \quad n = 1, 2, 3, \ldots$$

where $\omega = \sqrt{k/m}$, $\alpha = \sqrt{\mu\omega/\hbar}$, $N_n = \left(\frac{\alpha}{\sqrt{\pi}\, 2^n n!}\right)^{\frac{1}{2}}$ and $H_n(\xi)$ is the Hermite polynomial with the highest power n. We know that $H_n(\xi)$ is an even or odd function if $n$ is even or odd. If there is a particle of mass $m$ moving in the potential

$$U(x) = \begin{cases} \frac{1}{2} k x^2, & x \geq 0, \\ +\infty, & x < 0, \end{cases}$$

find the stationary wave functions and the energy levels of the particle.

3.(20pts) If we put 6 identical bosons or 6 identical spin-1/2 fermions into a one-dimensional infinite well of width $a$ (the energy levels for a particle of mass $m$ in the well is $E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$).

(a) Draw a schematic diagram to show how the energy levels are filled in each case when the system is in the ground state.

(b) Find the ground state energy of the system in each case.

(c) The atomic mass of a copper atom is 63.55; the density is 8.96 gcm$^{-3}$. Calculate the Fermi energy $E_F$ of the electron gas in copper. (Assume there is 1 valence electron per atom for copper.)

4. (20pts) Radioactive nuclei with decay constant $\lambda$ are produced in an accelerator at a constant rate $R_p$.

(a) Derive the differential equation for the number of radioactive nuclei *N*

(b) If the number of the nuclei is zero at $t = 0$, find the number *N* as a function of time *t*.

(c) $^{62}$Cu is produced at a rate 100 per second by placing ordinary copper($^{63}$Cu) in a beam of high-energy protons and $^{62}$Cu decays by beta-decay with a half-life of 10 min. How many $^{62}$Cu are there after the time long enough so that $dN/dt \to 0$?

5.(20pts) The empirical binding energy of a nucleus can be written as

$$E_b = a_1 A - a_2 A^{2/3} - a_3 Z^2 A^{-1/3} - a_4 (N - Z)^2 A^{-1}$$

(the pairing energy is not included), where $Z$ is the atomic number, $A$ is the mass number and $N$ is the neutron number of the nucleus. The values of the constants are

$a_1 = 15.8 MeV$, $a_2 = 17.8 MeV$, $a_3 = 0.71 MeV$, $a_4 = 23.7 MeV$.

(a) Determine the atomic number of the most stable nucleus for a given mass number $A$.

(b) Explain the physics meaning of the result.

6. (20pts) Determine if the following processes are mediated mainly by strong, electromagnetic or weak interactions or forbidden. Explain why.

(a) $\eta \to \pi^+ + \pi^- + \pi^0$.

(b) $\rho \to \pi^+ + \pi^-$

(c) $\pi^+ \to \mu^+ + \nu_e$

(d) $K^0 \to \pi^+ + \pi^-$

not required !!!

<u>Optional Problem</u> (20pts) Consider a spinless positive pion decaying at rest. (The initial angular momentum and linear momentum of the pion are zero.) A positive muon and a muon neutrino are emitted in opposite directions with equal and opposite momenta. Meanwhile, the two emitted particles have equal but opposite spins due to the angular momentum conservation.

(a) Draw a figure to show the process and the mirror image process of the decay.

(b) Can the mirror image process occur in nature? Why?

(c) What interaction governs the positive pion decay?

(d) What conclusion can you draw if the mirror process can/cannot occur in nature?

| 1 | 2 | 3 | 4 | 5 | 6 | Sum |
|---|---|---|---|---|---|---|
| /20 | /20 | /20 | /20 | /20 | /20 | /100 |

Name ________________________ Dec. 29, 2005 Thursday

Department __________________

SID _________________________

***University Physics***

Practice Final Examination

School of Intensive Instruction in Sciences and Arts, Nanjing University

Physical Constants

| | |
|---|---|
| Electron charge magnitude | $e = 1.60\times10^{-19}$ C |
| Dielectric constant of vacuum | $\varepsilon_0 = 8.85\times10^{-12}$ F/m |
| Permeability of vacuum | $\mu_0 = 4\pi\times10^{-7}$ N/A$^2$ |
| Vacuum speed of light | $c = 3.00\times10^{8}$ m/s |
| Electron rest mass | $m_e = 9.11\times10^{-31}$ kg $= 0.511$Mev/c$^2$ |
| Proton rest mass | $m_n = 1.672\times10^{-27}$ kg $= 938.3$MeV/c$^2$ |
| Neutron rest mass | $m_n = 1.675\times10^{-27}$ kg $= 939.6$MeV/c$^2$ |
| Atomic mass unit | 1 amu $= 1.660\times10^{-27}$ kg $= 931.5$MeV/c$^2$ |
| Electron volt (eV) | 1eV=$1.60\times10^{-19}$ J |
| Planck constant | $h = 6.63\times10^{-31}$ Js $\hbar = 2.11\times10^{-31}$ Js $= 0.66\times10^{-15}$ eVs |
| Boltzmann constant | $k_B = 1.38\times10^{-23}$ J/K $= 8.62\times10^{-5}$ eV/K |
| Bohr radius | $a_0 = 0.529$ Angstroms |

Select five out of following six problems.

1. (20pts). (a) Show that the fractional loss of energy of a photon during a Compton collision is given by

$$\frac{\Delta E}{E} = \frac{h\nu'}{m_e c^2}(1-\cos\phi)$$

where $\nu'$ is the frequency of scattered photon, $\phi$ is the angle between the momentum of incident photon and the momentum of scattered photon..

(b) Show that when a photon of frequency $\nu$ scatters from a free (rest) electron, the maximum kinetic energy of the electron is given by

$$K_{\max} = \frac{2(h\nu)^2}{2h\nu + m_e c^2}.$$

2.(20 pts) An electron is moving on a circular mesoscopic ring of perimeter $l$.
(a) Write the Schrödinger's equation of the electron and the boundary condition for the wave function of the electron.
(b) Obtain the energy levels and the corresponding stationary wave functions of the electron.
(c) Find the degeneracy of the energy levels.

3.(20pts) The asymmetry term of the empirical nuclear binding energy can be obtained through the Fermi gas model of the nucleus. The model is analogous to electron gas model of the metal. It assumes that the nucleus consists of independent proton and neutron gases. Consider a nucleus consisting of $N$ neutrons and $Z$ protons respectively, where $N + Z = A$, the mass number of the nucleus. The radius of the nucleus is given by $R = r_0 A^{1/3}$ and $r_0 = 1.2$ fm.
(a) Derive the expressions of Fermi energies for both neutron and proton gases.
(b) Find the total kinetic energy of the nucleons when the system is in its ground state.
(c) Show that the kinetic energy is minimized when $Z = N = A/2$.
(d) Find the difference of the kinetic energy from its minimum value when the proton number is deviated from $A/2$.

4.(20pts) The half-life of $^{235}$U is $7.04\times10^{8}\,y$.
(a) What is the decay constant of $^{235}$U?
(b) A sample of rock, which solidified with the earth $4.55\times10^{9}$ years ago, contains $N$ atoms of $^{235}$U. How many $^{235}$U atoms did the same rock have at the time it solidified?

5. The quark combinations of following particles are

$$K^{0}- d\bar{s} \qquad \pi^{-}- d\bar{u} \qquad K^{-}-s\bar{u} \qquad p - uud \qquad \Lambda^{0}-uds \quad \Sigma^{-}-dds\,.$$

(a) Find the charge number Q, the baryon number B and the strangeness S of these particles.
(b) Check the conservations of baryon number, charge and strangeness in the following processes.
(c) Determine if these processes are mediated by the strong or weak interaction, or are forbidden. Explain why.

(1) $\Lambda^{0} \to \pi^{-} + p$

(2) $\pi^{-} + p \to \Lambda^{0} + K^{0}$

(3) $K^{-} + p \to \Sigma^{-} + \pi^{+}$

(4) $K^{-} + p \to \Lambda^{0} + K^{0}$

Quantum numbers of three quarks

| Quark symbol | Charge number | Spin | Baryon number | Strangeness |
|---|---|---|---|---|
| u | 2/3 | 1/2 | 1/3 | 0 |
| d | −1/3 | 1/2 | 1/3 | 0 |
| s | −1/3 | 1/2 | 1/3 | −1 |

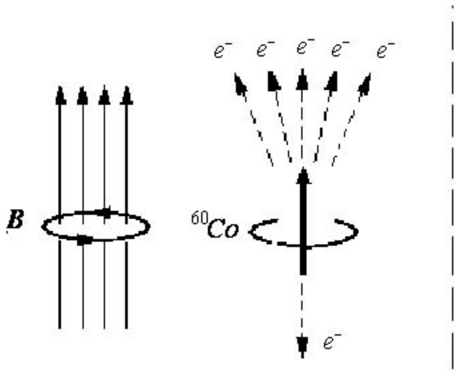
6.(20pts) In the $^{60}Co$ beta-decay experiment, the electrons emitted in the direction parallel to the applied magnetic field are found to be more than those emitted in the direction anti-parallel to the field. The original process is shown in the figure.

(a) Draw a figure to show the mirror image process of the beta-decay.

(b) Can the mirror image process occur in nature? Why?

(c) What conclusion can you draw if the mirror process can/cannot occur in nature?

Solution: (a)




Original process　　　　　　Mirror image process

| 1 | 2 | 3 | 4 | 5 | 6 | Sum |
|---|---|---|---|---|---|---|
| /20 | /20 | /20 | /20 | /20 | /20 | /100 |

Name ______________________ January 6th, 2008 Sunday

Department _________________

SID _______________________

***University Physics***

Final Examination

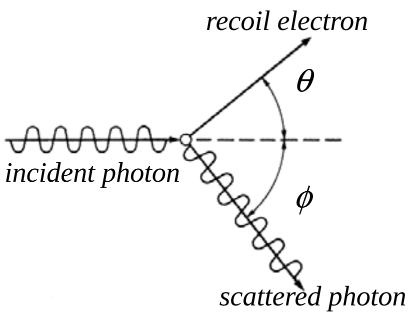
School of Intensive Instruction in Sciences and Arts, Nanjing University

Physical Constants

| | |
|---|---|
| Electron charge magnitude | $e = 1.60\times10^{-19}$ C |
| Dielectric constant of vacuum | $\varepsilon_0 = 8.85\times10^{-12}$ F/m |
| Permeability of vacuum | $\mu_0 = 4\pi\times10^{-7}$ N/A$^2$ |
| Vacuum speed of light | $c = 3.00\times10^{8}$ m/s |
| Electron rest mass | $m_e = 9.11\times10^{-31}$ kg = 0.511Mev/c$^2$ |
| Proton rest mass | $m_n = 1.672\times10^{-27}$ kg = 938.3MeV/c$^2$ |
| Neutron rest mass | $m_n = 1.675\times10^{-27}$ kg = 939.6MeV/c$^2$ |
| Atomic mass unit | 1 amu = $1.660\times10^{-27}$ kg = 931.5MeV/c$^2$ |
| Electron volt (eV) | 1eV=$1.60\times10^{-19}$ J |
| Planck constant | $h = 6.63\times10^{-34}$ Js $\hbar = 1.05\times10^{-34}$ Js = $6.58\times10^{-16}$ eVs |
| Boltzmann constant | $k_B = 1.38\times10^{-23}$ J/K = $8.62\times10^{-5}$ eV/K |
| Bohr radius | $a_0 = 0.529$ Angstroms |

Select five out of following six problems.

1. (20pts) In Compton Scattering, an incoming X-ray photon of frequency $\nu$ scatters off an electron that is initially at rest. The direction of scattered photon makes an angle $\phi$ to the direction of the incident X-ray, and the electron gains energy and recoils in the direction that makes an angle $\theta$ to the direction of the incident photon as shown in the figure.




(a) Show that the increase of the wavelength of the scattered photon is given by

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c}(1-\cos\phi).$$

(b) Show that the scattering angle of the electron $\theta$ is given by

$$\cot\theta = \left(1+\frac{h\nu}{m_e c^2}\right)\tan\frac{\phi}{2}.$$

2. (20 pts) An electron is moving in a one dimensional infinite potential well

$$V(x) = \begin{cases} 0 & 0 \le x \le a \\ +\infty & \text{elsewhere} \end{cases}.$$

(a) Write the Schrödinger's equation of the electron

(b) Obtain the energy levels and the corresponding stationary wave functions of the electron.

(c) Show that the expectation (average) value of the kinetic energy of a particle in one dimension can be expressed as

$$\langle \hat{T} \rangle \equiv \left\langle \frac{\hat{p}^2}{2m} \right\rangle = \frac{\hbar^2}{2m} \int \left| \frac{d\psi}{dx} \right|^2 dx$$

where $\psi(x)$ is the normalized wave function and satisfies $\lim_{x\to\pm\infty} \psi(x) = 0$ (bound state).

3. (20pts) The energy levels and the corresponding stationary wave functions of a particle in the one-dimensional harmonic potential $V(x) = 1/2\, kx^2$ can be given as

$$E_n = (n + 1/2)\hbar\omega\,, \quad \psi_n(x) = N_n e^{-\frac{\alpha^2 x^2}{2}} H_n(\alpha x), \quad n = 0,1,2,\cdots.$$

If we put 8 identical fermions of spin 1/2 or 8 identical bosons of spin 0 in the harmonic potential,

(a) Draw a schematic diagram to show how the energy levels are filled in each case when the system is in the ground state.

(b) Find the ground state energy of the system in each case.

4.(20pts) An unstable element of half-life $T_{1/2}$ is produced in a nuclear reactor at a constant rate $R$.
(a) What is the equilibrium quantity of the element?
(b) How much time is required to produce $50\%$ of the equilibrium quantity of the element?
(c) In terms of the rest mass of the parent atom $M_P$ and the rest mass of the daughter atom $M_D$, determine the Q-values for $\beta^-$ decay, $\beta^+$ decay and electron capture (K-shell capture):

$$ {}^{A}_{Z}\mathrm{P} \to {}^{A}_{Z+1}\mathrm{D} + e^- + \bar{\nu}_e \quad (\beta^- \text{ decay}) $$

$$ {}^{A}_{Z}\mathrm{P} \to {}^{A}_{Z-1}\mathrm{D} + e^+ + \nu_e \quad (\beta^+ \text{ decay}) $$

$$ {}^{A}_{Z}\mathrm{P} + e^- \to {}^{A}_{Z-1}\mathrm{D} + \nu_e \quad (\text{electron capture}). $$

5. The quark combinations of following particles are

$$ \pi^- - d\bar{u} \qquad K^0 - d\bar{s} \qquad K^- - s\bar{u} \qquad p - uud \qquad \Xi^0 - uss \qquad \Omega^- - sss . $$

(a) Find the charge number Q, the baryon number B and the strangeness S of these particles.
(b) Determine if the following processes are mainly mediated by the strong or weak interaction, or are forbidden. Where a process is forbidden, give all of the reasons why this is the case.

(1) $K^- + p \to \Omega^- + K^+ + K^0$

(2) $\Omega^- \to \Xi^0 + \pi^0$

(3) $K^0 \to \pi^- + \mu^+ + \nu_\mu$

(4) $p \to e^+ + \gamma$

*Quantum numbers of three quarks*

| Quark symbol | Charge number | Spin | Baryon number | Strangeness |
|---|---|---|---|---|
| u | $2/3$ | $1/2$ | $1/3$ | 0 |
| d | $-1/3$ | $1/2$ | $1/3$ | 0 |
| s | $-1/3$ | $1/2$ | $1/3$ | $-1$ |

Solution:

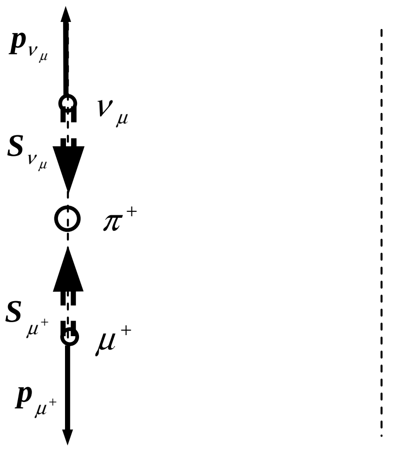
| Particle | Q | B | S | Particle | Q | B | S |
|---|---|---|---|---|---|---|---|
| $\pi^-$ | | | | p | | | |
| $K^0$ | | | | $\Xi^0$ | | | |
| $K^-$ | | | | $\Omega^-$ | | | |

6.(20pts) Consider a spinless positive pion decaying at rest. (The initial momentum of the pion is zero.) A positive muon and a muon neutrino are emitted in opposite directions with equal and opposite momenta. Meanwhile, the two emitted particles have equal but opposite spins due to the angular momentum conservation.

(a) Draw a figure to show the charge-conjugate process.

(b) Can the charge-conjugate process occur in nature? Why?

(c) What conclusion can you make if the charge-conjugate process can/cannot occur in nature?




Original process　　　　　　　　Charge-conjugate process

| 1 | 2 | 3 | 4 | 5 | 6 | sum |
|---|---|---|---|---|---|---|
| /20 | /20 | /20 | /20 | /20 | /20 | /100 |

Jan. 11 2014, Saturday

Name______________________________

Department_________________________ SID___________________________

***University Physics***

Final Examination

Kuang Yaming Honors School, Nanjing University

Some physical constants: $h = 6.63\times10^{-34}\,\mathrm{J\cdot s}$, $1\mathrm{eV} = 1.60\times10^{-19}\,\mathrm{J}$ , $m_e = 9.11\times10^{-31}\,\mathrm{kg} = 0.511\,\mathrm{MeV/c^2}$, $1\,\mathrm{u} = 1.660\times10^{-27}\,\mathrm{kg} = 931.5\,\mathrm{MeV/c^2}$, $hc = 1.24\times10^3\,\mathrm{eV\cdot nm}$ .

<u>Select five out of the following six problems.</u>

1. A $\pi^0$ meson at rest decays into two photons ($\pi^0 \to \gamma + \gamma$ ).

(a) What type of interaction is mainly responsible for the decay?

(b) What are the energy and momentum of each photon?

(c) What is the wavelength $\lambda$ of the photons? (The mass of the $\pi^0$ is 135 MeV/$c^2$.)

2. Consider a one-dimensional simple harmonic oscillator of mass $m$ and natural circular frequency $\omega$.

(a) Write down the Hamiltonian $\hat{H}$ of the oscillator.

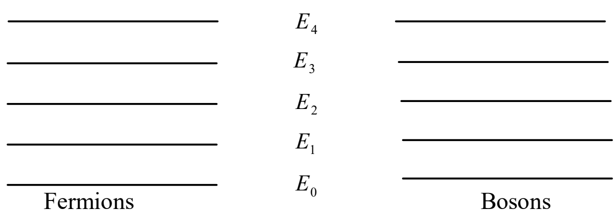
(b) Show that the wave function $\psi(x) = Nx\exp\left(-\frac{\alpha^2 x^2}{2}\right)$, where $\alpha = \sqrt{\frac{m\omega}{\hbar}}$ and $N$ is a constant, is an eigenfunction of the Hamiltonian $\hat{H}$ of the oscillator.

(c) Find $\langle x\rangle$, the mean value of the position of the particle, if the particle is in the state described by the wave function in (b).

3. (a) The energy levels a particle in the one-dimensional harmonic potential $V(x) = \frac{1}{2} m\omega^2 x^2$ can be given as $E_n = (n + \frac{1}{2})\hbar\omega$, $n = 0,1,2,\cdots$. If 8 identical fermions of spin 1/2 or 8 identical bosons of spin 0 are put in the harmonic potential, how the energy levels are filled in each case when the system is in the ground state? (Draw on the figure, $\uparrow$ as an up-spin fermion, $\downarrow$ as a down-spin fermion, $\circ$ as a boson. ) What is the ground state energy of the system in each case?

(b) The Fermi energy $E_F$ for the electrons in copper is measured to be 7.00 eV. If the mass of the electron is $m = 0.511\,\mathrm{MeV/c^2}$, find the volume density $n$ of the conducting electrons in copper.

Solution (a)



4. (a) An unstable element of half-life $T_{1/2}$ is produced in a nuclear reactor at a constant rate $R$. What is the equilibrium quantity of the element? How much time is required to produce $50\%$ of the equilibrium quantity of the element?

(b) In terms of the rest mass of the parent atom $M_P$ and the rest mass of the daughter atom $M_D$, determine the Q-values for $\beta^-$ decay, $\beta^+$ decay and electron capture (K-shell capture):

$$ {}^{A}_{Z}\mathrm{P} \to {}^{A}_{Z+1}\mathrm{D} + e^- + \bar{\nu}_e \quad (\beta^- \text{ decay}) $$

$$ {}^{A}_{Z}\mathrm{P} \to {}^{A}_{Z-1}\mathrm{D} + e^+ + \nu_e \quad (\beta^+ \text{ decay}) $$

$$ {}^{A}_{Z}\mathrm{P} + e^- \to {}^{A}_{Z-1}\mathrm{D} + \nu_e \quad (\text{electron capture}). $$

5. The quark combinations of following particles are

$$ \pi^- - d\bar{u} \quad \pi^+ - u\bar{d} \quad K^0 - d\bar{s} \quad K^+ - u\bar{s} \quad K^- - s\bar{u} \quad p - uud \quad \Xi^0 - uss \quad \Omega^- - sss. $$

(a) Find the charge number Q, the baryon number B and the strangeness S of these particles.

(b) Determine if the following reactions are mainly mediated by the strong or weak interaction, or are forbidden. Where a reaction is forbidden, give the reason.

(1) $K^- + p \to \Omega^- + K^+ + K^0$

(2) $\Omega^- \to \Xi^0 + \pi^0$

(3) $K^0 \to \pi^- + \mu^+ + \nu_\mu$

(4) $p \to e^+ + \gamma$

*Quantum numbers of three quarks*

| Quark symbol | Charge number | Spin | Baryon number | Strangeness |
|---|---|---|---|---|
| u | $2/3$ | $1/2$ | $1/3$ | 0 |
| d | $-1/3$ | $1/2$ | $1/3$ | 0 |
| s | $-1/3$ | $1/2$ | $1/3$ | $-1$ |

Solution: (a)

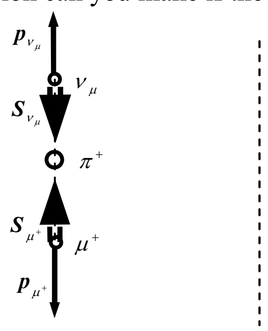
| *Particle* | Q | B | S | *Particle* | Q | B | S |
|---|---|---|---|---|---|---|---|
| $\pi^-$ | | | | $\pi^+$ | | | |
| $K^0$ | | | | $K^+$ | | | |
| $K^-$ | | | | p | | | |
| $\Xi^0$ | | | | $\Omega^-$ | | | |

6. Consider a spinless positive pion decaying at rest. (The initial angular momentum and linear momentum of the pion are zero.) A positive muon and a muon neutrino are emitted in opposite directions with equal and opposite momenta $\boldsymbol{p}_{\mu^+}$ and $\boldsymbol{p}_{\nu_\mu}$. Meanwhile, the two emitted particles have equal but opposite spins ($\boldsymbol{S}_{\mu^+}$ and $\boldsymbol{S}_{\nu_\mu}$) due to the angular momentum conservation.

(a) Draw a figure to show the charge-conjugate process.

(b) Can the charge-conjugate process occur in nature? Why?

(c) What conclusion can you make if the charge-conjugate process can/cannot occur in nature?




Original process Charge-conjugate process

| 1 | 2 | 3 | 4 | 5 | 6 | Sum |
|---|---|---|---|---|---|---|
| /20 | /20 | /20 | /20 | /20 | /20 | /100 |

Name____________________________ Jan 15, 2016 Friday

SID______________________________ School/Department______________________

# ***University Physics***

## Final Examination

## Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu)=\int_0^{\infty} e^{-t}t^{\nu-1}dt,\ \ \Gamma(\nu+1)=\nu\Gamma(\nu),\ \ \Gamma(1)=1,\ \ \Gamma(1/2)=\sqrt{\pi}\ .$$

| c | $h$ | $e$ | $m_e$ | u | $N_A$ | $k_B$ |
|---|---|---|---|---|---|---|
| $3.0\times10^{8}$ m/s | $6.63\times10^{-34}$ J·s<br>$1.24\times10^{3}$eV ·nm/c | $1.6\times10^{-19}$ C | $9.11\times10^{-31}$ kg<br>0.511MeV/$c^2$ | 931.494MeV/$c^2$ | $6.02\times10^{23}$ mol$^{-1}$ | $8.617\times10^{-5}$eV/K |

The quark combinations of some particles: p—uud n—udd $\Lambda^0$ – uds $\Sigma^+$ – uus

Quantum numbers of three quarks:

| Quark symbols | Charge numbers | Spin | Baryon numbers | Strangeness |
|---|---|---|---|---|
| u | 2/3 | 1/2 | 1/3 | 0 |
| d | −1/3 | 1/2 | 1/3 | 0 |
| s | −1/3 | 1/2 | 1/3 | −1 |

Select five out of the following six problems.

1.(20 pts.) When neutrons are in thermal equilibrium within a reactor, they satisfy the equipartition theorem.

(a) If the temperature of the reactor is 300 K, what is the average kinetic energy $\bar{\varepsilon}$ of such a thermal neutron?

(b) The mass of neutron is known to be 1.0087 u, What is the corresponding de Broglie wavelength $\lambda$?

2. (20 pts.) After long effort, in 1902, Marie and Pierre Curie succeeded in separating from uranium ore the first substantial quantity of radium, one decigram of pure $RaCl_2$(radium chloride). Actually the radium they obtained was the radioactive isotope $^{226}$Ra, which has a half-life of 1600 a.

(a) How many nuclei of $^{226}$Ra had they isolated?

(b) What is the average life time $\tau$ of the isotope $^{226}$Ra.

(c) What is the decay rate $R$ of their sample at a late time $t$?

(The atomic mass of Cl(chlorine) is 35.5u .)

3.(20 pts.) An electron of a stationary hydrogen atom passes from the fifth level to the ground state, with a photon emitted in the process.

(a) Calculate the energy $E_1$ of the ground state and the energy $E_5$ of the fifth level, according to the Bohr theory.

(b) Considering the recoil of the hydrogen atom, what is the energy $E_\gamma$ of the emitted photon, to the order of $(E_5 - E_1)^2 / mc^2$ where $m = 938.782\,\mathrm{MeV}/c^2$ is the rest mass of the hydrogen atom?

(c) To the same order of magnitude, what is the recoil energy $E_{recoil}$ of the hydrogen atom?

4.(20 pts.) Black body radiation can be treated as an ideal gas of photons. As such, its density of states $D(\nu)$ in the frequency $\nu$ space is $D(\nu) = V8\pi\nu^2c^{-3}$ and the number of quantum states within the interval of frequency $\nu$ to $\nu + d\nu$ is $V8\pi\nu^2c^{-3}d\nu$, where $V$ is the volume of the photon gas.

(a) Show that the internal energy of the photon gas is $U = \alpha T^4 V$ where $T$ is temperature of the system and $\alpha = k_B^4\Gamma(4)\varsigma(4)/\pi^2c^3\hbar^3$ .

(b) Find the heat capacity at constant volume $C_V$ of the system.

(c) Find the Helmholtz free energy $F$ and the Gibbs free energy $G$ of the system.

(The fundamental equation of thermodynamics here can be written as $dU = TdS - pdV$ .

$\int_0^{+\infty} \frac{x^{\nu-1}}{e^x - 1} dx = \Gamma(\nu)\varsigma(\nu), \nu > 0.$ )

5. (20 pts.) A one-dimensional simple harmonic oscillator with mass $m$ and natural angular frequency $\omega$ is found to stay in the quantum state of following normalized wave function

$$\psi = c\left(\frac{\alpha}{\sqrt{\pi}}\right)^{1/2} e^{-\frac{1}{2}\alpha^2x^2} + \left(\frac{\alpha}{\sqrt{\pi}}\right)^{1/2} \alpha x e^{-\frac{1}{2}\alpha^2x^2},$$

where $\alpha = (m\omega/\hbar)^{1/2}$, the $c$ is a positive constant to be determined.

(a) Determine the constant $c$.

(b) Is the $\psi$ an eigenfunction of the Hamiltonian of the oscillator?

(c) Find $\langle H \rangle$, the average energy of the oscillator.

(d) Find $\Delta E$, the uncertainty or fluctuation of the energy of the oscillator.

6. (20 pts.) For each of the following reactions state whether or not it is forbidden. If it is forbidden, state a conservation law that is violated.

(a) $\Lambda^0 \to p + \pi^0$

(b) $\overline{p} + p \to \mu^+ + e^-$

(c) $n \to p + e^- + \nu_e$

(d) $n + \nu_e \to p + e^-$

(e) $\Sigma^+ \to \Lambda^0 + \pi^+$

| 1 | 2 | 3 | 4 | 5 | 6 | Sum |
|---|---|---|---|---|---|---|
| /20 | /20 | /20 | /20 | /20 | /20 | /100 |

Name____________________________ Jan 3, 2017 Tuesday

SID______________________________ School/Department____________________

***University Physics***

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu)=\int_0^{\infty} e^{-t}t^{\nu-1}dt,\ \Gamma(\nu+1)=\nu\Gamma(\nu),\ \Gamma(1)=1,\ \Gamma(1/2)=\sqrt{\pi}.$$

| c | $h$ | $\hbar$ | $e$ | $m_e$ | u | $N_A$ | $k_B$ |
|---|---|---|---|---|---|---|---|
| $3.0\times10^{8}$ m/s | $6.63\times10^{-34}$ J·s<br>$1.24\times10^{3}$eV ·nm/c | $1.05\times10^{-34}$ J·s | $1.6\times10^{-19}$ C | $9.11\times10^{-31}$ kg<br>0.511MeV/$c^2$ | 931.494MeV/$c^2$ | $6.02\times10^{23}$ mol$^{-1}$ | $8.617\times10^{-5}$eV/K |

The quark combinations of some hadrons: $\mathrm{p}-\mathrm{uud}$ $\mathrm{n}-\mathrm{udd}$ $\Lambda^0-\mathrm{uds}$ $\pi^+-\mathrm{u\overline{d}}$ $\pi^- -\mathrm{d\overline{u}}$ $\mathrm{K}^+-\mathrm{u\overline{s}}$ $\mathrm{K}^0-\mathrm{d\overline{s}}$

Quantum numbers of three quarks:

| Quark symbols | Charge numbers | Spin | Baryon numbers | Strangeness |
|---|---|---|---|---|
| u | 2/3 | 1/2 | 1/3 | 0 |
| d | −1/3 | 1/2 | 1/3 | 0 |
| s | −1/3 | 1/2 | 1/3 | −1 |

Select five out of the following six problems.

1.(20 pts.) A $^{57}$Fe nucleus in the first excited state decays to its ground state with the emission of a $\gamma$ photon of 14.4 keV in a mean lifetime of 141 ns.

(a) Estimate the nuclear energy level width $\Delta E$ of the first excited state.

(b) What is the recoil kinetic energy $E_K$ of an initially stationary $^{57}$Fe nucleus when it emits a $\gamma$ photon of 14.4 keV?

2. (20 pts.) The plutonium $^{239}$Pu is a by-product in nuclear reactor. It is radioactive, decaying by alpha decay with a half-life of $2.4\times10^{4}$ a (1 a = $\pi\times10^{7}$ s). Plutonium is also one of the most toxic chemicals known and its lethal dose to human is as little as 2 mg.

(a) How many nuclei does it have in the chemically lethal dose of $^{239}$Pu?

(b) What is the decay rate ( or activity ) $R$ of the amount of $^{239}$Pu nuclei in (a) exactly after a time period of $7.2\times10^{4}$ a?

3.(20 pts.) (a) For the particles in the first column of the following table, classify the particles as lepton, baryon and meson, find the charge number, lepton number, baryon number and strangeness of them and write the results into the table.

| Particles | Classification lepton/baryon/meson | Lepton number | | | Baryon number | Charge number | Strangeness |
|---|---|---|---|---|---|---|---|
| | | $L_e$ | $L_\mu$ | $L_\tau$ | | | |
| $e^+$ | | | | | | | |
| $K^0$ | | | | | | | |
| $\Lambda^0$ | | | | | | | |
| $\nu_\mu$ | | | | | | | |
| p | | | | | | | |
| $\pi^-$ | | | | | | | |

(b) For the following reactions or decays, indicate if they are allowed or forbidden. Where it is forbidden, give a reason.

| Reactions or decays | Allowed/Forbidden | Reason |
|---|---|---|
| $\mu^- \rightarrow e^- + \nu_\mu + \bar{\nu}_e$ | | |
| $p + p \rightarrow p + n + \pi^+$ | | |
| $K^0 \rightarrow \pi^- + \mu^+ + \nu_\mu$ | | |
| $n \rightarrow e^+ + e^- + \nu_e$ | | |
| $K^- + p \rightarrow \Lambda^0 + K^0$ | | |

4. (20 pts.) A particle of mass $m$ is moving in a one-dimensional simple harmonic potential and its natural circular frequency is $\omega$. The particle is in a stationary quantum state of the normalized wave function

$$\psi(x) = Cx\mathrm{e}^{-\frac{1}{2}\alpha^2 x^2}$$

where $\alpha = \sqrt{m\omega/\hbar}$ and $C$ is a positive constant to be determined.

(a) Determine the constant $C$.

(b) Find the most probable position(s) $x_{mp}$ of the particle.

(c) Find the probability flux density $J$ of the particle.

(Hint: The probability flux density is given by $\boldsymbol{J} = \frac{\hbar}{2im}\left(\psi^* \nabla \psi - \psi \nabla \psi^*\right)$ where $\psi$ is a normalized wave function.)

5. (20 pts.) A particle of mass $m$ moves in a one-dimensional interval [0, $a$] on the x-axis and the potential energy of the particle is identically zero within the interval. The wave function of the particle is assumed to satisfy the periodic boundary condition $\psi(0) = \psi(a)$.

(a) Show that the following states are eigenfunctions of both the momentum $\hat{p}$ and Hamiltonian $\hat{H}$.

$$\psi_n(x) = \begin{cases} \frac{1}{\sqrt{a}} e^{ik_n x}, & 0 \le x \le a \\ 0\ , & \text{elsewhere,} \end{cases} \quad \text{where } k_n = \frac{2\pi n}{a},\ n = 0, \pm 1, \pm 2, \cdots.$$

(b)Show that the eigenfunctions in (a) are orthonormalized, i.e. ,

$$\int_0^a \psi_m^*(x)\psi_n(x)dx = \delta_{mn}, \quad \text{where } \delta_{mn} = \begin{cases} 1, & m = n, \\ 0, & m \ne n \end{cases} \text{ is the Kronecker delta.}$$

(c) If the particle is in a quantum state that is a linear combination of the eigenfunctions in (a), say

$$\psi(x) = \frac{1}{2}\psi_{-1}(x) + \frac{1}{2}\psi_1(x) + c\psi_5(x)$$

where $c$ is a positive constant and $\psi(x)$ is normalized, find the constant $c$.

(d) Find the average momentum $\langle p \rangle$ and the uncertainty of momentum $\Delta p$ if the particle is in the quantum state in (c). .

6. (20 pts.) The rest mass of the electron neutrino $\nu_e$ is nearly zero so that its energy- momentum relation can be written as $E = pc$ where $c$ is the speed of light in free space. Suppose that the particle number density of $\nu_e$ in a space is $n$ and the interaction between the electron neutrinos is so small that the neutrinos can be taken as a non-interacting Fermi gas.

(a) Find the Fermi wave number $k_F$ of the neutrino gas.

(b) Find the Fermi energy $E_F$ Fermi temperature $T_F$ of the neutrino gas.

(c) Find the average energy per particle $\overline{E}$ of the electron neutrinos under zero temperature.